In [6]:
```
#import library
import pandas as pd
#creating a series
marks=pd.Series([78,85,90,66,72,88,91,55,60,71,74,88,45,58,79,84,82,69,88,9245,5
#---------------------------------------------------------------------------------
#display first 5 values
print("First 5 elements: ")
print(marks.head(5))
print("----------------------------------")
print("----------------------------------")
#to display last 7 elements
print("Last 7 values:")
print(marks.tail(7))
print("----------------------------------")
print("----------------------------------")
#to display the frequency of each values
print("Frequency of each values")
print(marks.value_counts())
print("----------------------------------")
print("----------------------------------")
#to display unique values
print("Unique vlues:")
print(marks.unique)
print("----------------------------------")
print("----------------------------------")
#to show total unique values
print("Total unique values:")
print("Total unique values :",marks.nunique())
```

```
First 5 elements:
0    78
1    85
2    90
3    66
4    72
dtype: int64
----------------------------------
----------------------------------
Last 7 values:
20    57
21    28
22    48
23    59
24    28
25    59
26    68
dtype: int64
----------------------------------
----------------------------------
Frequency of each values
88      3
28      2
59      2
78      1
85      1
90      1
66      1
72      1
91      1
55      1
60      1
71      1
74      1
45      1
58      1
79      1
84      1
82      1
69      1
9245    1
57      1
48      1
68      1
Name: count, dtype: int64
----------------------------------
----------------------------------
Unique vlues:
<bound method Series.unique of 0       78
1       85
2       90
3       66
4       72
5       88
6       91
7       55
8       60
9       71
10      74
11      88
```

```
12      45
13      58
14      79
15      84
16      82
17      69
18      88
19    9245
20      57
21      28
22      48
23      59
24      28
25      59
26      68
dtype: int64>
----------------------------------
----------------------------------
Total unique values:
Total unique values : 23
```

In [9]:
```
#to create a list
data=pd.Series([78,90,89,90,45],index=['Hindi','English','Science','Maths','Comp
# to sort the data as per values
print(data.sort_values())
#to sort data as per index
print("----------------------------------")
print("Sorted data as per index:")
print(data.sort_index())
```

```
Computer    45
Hindi       78
Science     89
English     90
Maths       90
dtype: int64
----------------------------------
Sorted data as per index:
Computer    45
English     90
Hindi       78
Maths       90
Science     89
dtype: int64
```

In [11]:
```
print("----------------------------------")
#to sort the null data
data=pd.Series([50,60,70,None,68,83,None])
print(data.isnull())
print("----------------------------------")
print("----------------------------------")
```

```
---------------------------------
0    False
1    False
2    False
3     True
4    False
5    False
6     True
dtype: bool
---------------------------------
---------------------------------
```

In [13]:
```
import numpy as np
data1=pd.Series([50,67,89,None,89,56,80,None,86])
#to fill none values as 0
print(data1.fillna(0))
```

```
0    50.0
1    67.0
2    89.0
3     0.0
4    89.0
5    56.0
6    80.0
7     0.0
8    86.0
dtype: float64
```

In [14]:
```
data2=pd.Series([50,67,89,None,89,56,80,None,86])
#missing value treatment as filling data with average value
print(data2.fillna(data2.mean()))
```

```
0    50.000000
1    67.000000
2    89.000000
3    73.857143
4    89.000000
5    56.000000
6    80.000000
7    73.857143
8    86.000000
dtype: float64
```

In [15]:
```
data3=pd.Series([50,67,89,None,89,56,80,None,86])
#dropping missing values
print(data3.dropna())
```

```
0    50.0
1    67.0
2    89.0
4    89.0
5    56.0
6    80.0
8    86.0
dtype: float64
```

In [16]:
```
#to create a series
marks=pd.Series([90,80,70,89,60])
#-------------------------------------
#to display the series
print(marks)
```

```
#-------------------------------------
print("=============================")
data=[80,90,70,60,50]
#indexing
subname=['Science','maths','english','Computer','Hindi']
markss=pd.Series(data,index=subname)
#displaying
print(markss)
```

```
0    90
1    80
2    70
3    89
4    60
dtype: int64
=============================
Science     80
maths       90
english     70
Computer    60
Hindi       50
dtype: int64
```

In [ ]: